

R&D Proposal for Mushroom Direct

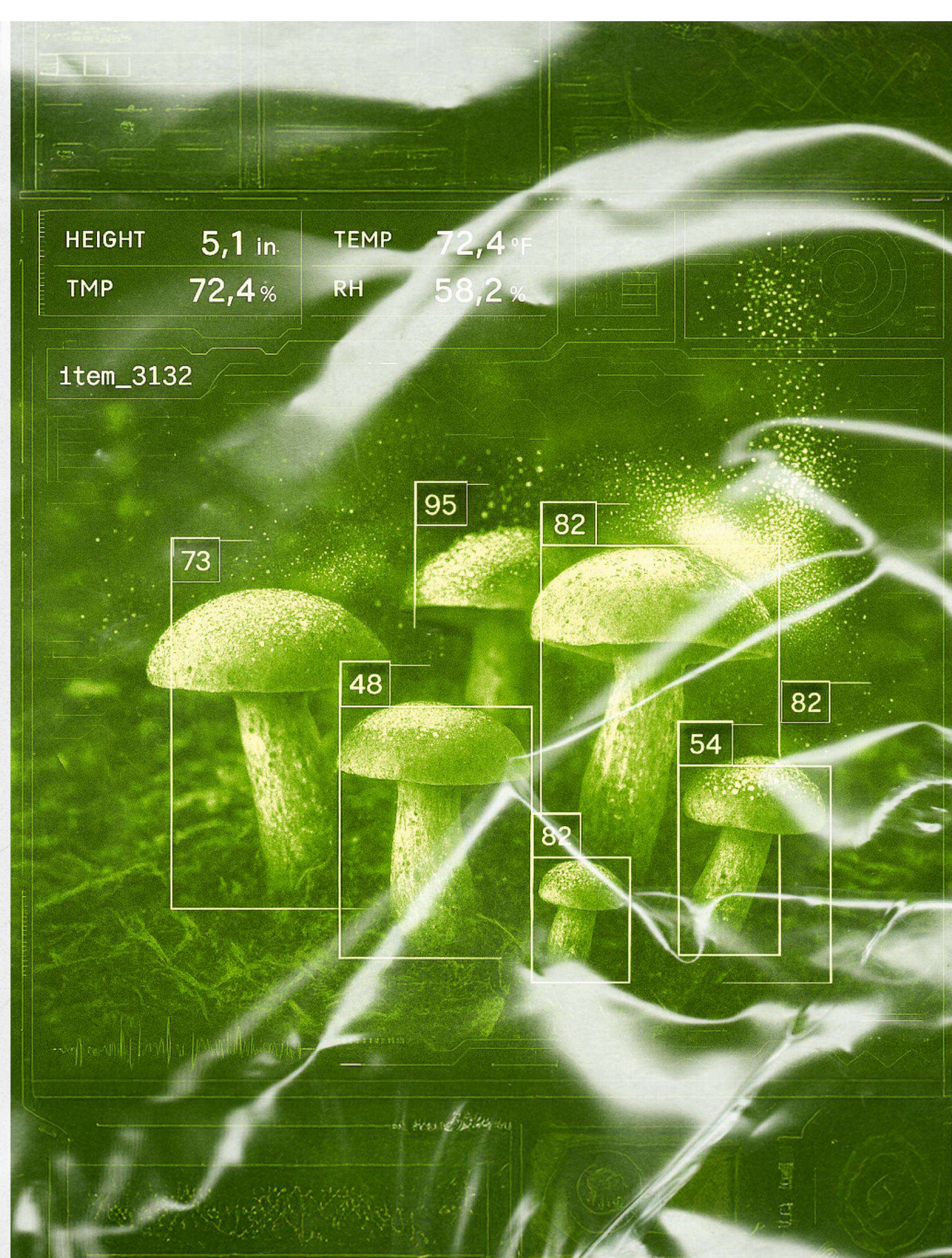
Computer Vision *for Mushroom Growth Analysis*

Growth stage recognition
sporulation, active growth, maturity

Quality assessment
shape, color, density, defect detection

Volume estimation
mass and marketable yield prediction

Support for up to 20 mushroom species
including wood-based and exotic varieties



Inventory Prediction & Management Algorithms



Yield and harvest timing forecasting

Logistics and storage optimization

Automatic stock and availability updates for warehouse and retail

Predictive models based on growth trends and environmental conditions

Investor & Partner Showcase Website

Modern landing page highlighting product innovation

Live production data feed
growth monitoring, AI analytics

Interactive lifecycle visualization from mushroom to product

E-Commerce Platform *Integrated with the Farm*

Real-time sync with computer vision.
Only available stock displayed

Pre-order mode based on growth forecasts

Logistics integration, readiness notifications, B2B API support

Wholesale and retail order management



Robotics & Growing Conditions Optimization

Robotic harvesting linked to computer vision

Data-driven climate control adjustments:

- Temperature
- Humidity
- Lighting
- CO₂ levels

Automated optimization of conditions per mushroom species

Goal: increased size, quality, and yield!

